

YUE (MELODY) HU

AI Product Engineer — 9 years shipping interactive & AI products end-to-end

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Engineer who owns the product: from ambiguous brief to shipped system. Built the agentic AI architecture behind

Create with AI — TikTok Effect House's AI generation tool that turns effect ideas into real AR effects, live on TikTok.

Client-facing agency background: IBM, Hershey, Santander (Webby Award). U.S. work authorization (no sponsorship).

EXPERIENCE

Senior Research Engineer · ByteDance / TikTok

Jan 2024 – Present

- **Architected AIGE, the AI system behind “Create with AI”** — type an effect idea, get a runnable effect live on TikTok. Lead the end-to-end agent architecture — **LLM classifies, decomposes, and plans against validated schemas; deterministic code executes** — holding quality across body tracking, voice recognition, and game mechanics.
- **Designed AIGE’s knowledge & skill retrieval (RAG)** — hierarchical knowledge (sceneObject → component → property) grounding MCP tool calls into Effect House; retrievable skill packs per game / effect genre.
- **Built AIGE’s evaluation system** — designed the benchmark and built automated validation of generated effects to measure and drive end-to-end generation success.

Interactive Engineer · ByteDance / TikTok Effect House

Jan 2021 – Jan 2024

- **Designed & shipped the Material Graph node system** of the Effect House creation tool — defined the node taxonomy (Transition / 2D SDF / Effect), specced inputs & outputs, and built nodes from prototype to production.
- **Led the Template production team** — shipped 120+ templates, driving **over 40% of all TikTok effect creations**.
- **Led Effect House community education** — 50+ tutorial videos and regular YouTube live sessions; scaled to **400k+ Discord members and 64k YouTube subscribers**.

Creative Technologist · Havas Worldwide, New York

Jun 2017 – Dec 2020

- Client-facing delivery across 4 years of agency work: translated ambiguous briefs from **IBM, Hershey, VUSE, Santander, Adidas, Hugo Boss** into shipped AR/VR/WebGL products — owning storyboards, architecture, and production code.
- **Webby Award winner (2019, Augmented Reality)** — “In Someone Else’s Shoes” for Santander.
- Lead creative technologist on **IBM Watson Holobot** (IBM Think 2018) and **Hershey’s Pop Kiss Studio** VR retail installation — interaction systems, particle VFX, and on-site user testing with real customers.

Adjunct Faculty · Parsons School of Design

Jan – May 2019

- Taught **“Warping Reality”** — AR development with Unity, ARKit/ARCore, Vuforia.

SELECTED PROJECT

Sound Visualization — Agentic VFX System for Live Music · solo · in progress

2026

- An agentic system that turns a song into a live, editable visual performance: **three agent stages with typed, human-editable artifacts** (music analysis → visual plan → performance score), WebGL runtime with live mic modulation, built for live stage projection.
- Genetic VFX engine: prompt → 4 candidates → mutate / breed → lock per segment. **Agents propose, humans commit**. Case study: huyue.space/work/sound-viz

SKILLS

AI systems Agentic architecture & multi-agent pipelines, LLM planning, function calling & MCP, structured outputs, RAG & skill libraries, LLM/VLM evals & benchmarks, AI-assisted dev (Claude Code, Cursor)

Engineering JavaScript/TypeScript, GLSL / WebGL / Three.js, C# / Unity3D, Python, React, Node, audio DSP

Interaction & visual craft Interaction design, motion design, real-time visual aesthetics (shaders, particles), Figma

EDUCATION

New York University — M.P.S., Interactive Telecommunications Program (ITP) · Red Burns Scholarship

2015 – 2017

Southeast University — B.Eng., Biomedical Engineering · Outstanding Graduate

2009 – 2013